

$$\begin{array}{ll}
 2 \cdot (1+1) = 2 \cdot 1 = 2 & 2 \cdot (1+2) = 2 \cdot 1 = 2 \\
 2 \cdot 1 + 2 \cdot 1 = 2+2=2 & 2 \cdot 1 + 2 \cdot 2 = 2+2=2 \\
 2 \cdot (2+0) = 2 \cdot 0 = 2 & 2 \cdot (2+1) = 2 \cdot 1 = 2 \\
 2 \cdot 2 + 2 \cdot 0 = 2+2=2 & 2 \cdot 2 + 2 \cdot 1 = 2+2=2
 \end{array}$$

$\therefore$ The distributive law $x \cdot (y+z) = x \cdot y + x \cdot z$ holds.

The inverse of $+$ doesn't exist. There does not exist an element y such that $0+y=2$.

The inverse of $\cdot$ doesn't exist. There does not exist an element y such that $1 \cdot y = 0$.

2-3: Demonstrate by means of truth tables the validity of the following theorems of Boolean algebra

(a) The associative laws

x	y	z	$x+y$	$(x+y)+z$	$y+z$	$x+(y+z)$
0	0	0	0	0	0	0
0	0	1	0	1	1	1
0	1	0	1	1	1	1
0	1	1	1	1	0	0
1	0	0	1	1	0	0
1	0	1	1	0	1	1
1	1	0	0	0	1	1
1	1	1	0	0	0	0

x	y	z	$x \cdot y$	$(x \cdot y) \cdot z$	$y \cdot z$	$x \cdot (y \cdot z)$
0	0	0	0	0	0	0
0	0	1	0	0	0	0
0	1	0	0	0	0	0
0	1	1	0	0	1	0
1	0	0	0	0	0	0
1	0	1	0	0	0	0
1	1	0	1	0	0	0
1	1	1	1	1	1	1

(b) De Morgan's theorems for three variables

x	y	z	$x+y+z$	$(x+y+z)'$	x'	y'	z'	$x' \cdot y' \cdot z'$
0	0	0	0	1	1	1	1	1
0	0	1	1	0	1	1	0	0
0	1	0	1	0	1	0	1	0
0	1	1	0	1	1	0	0	0
1	0	0	1	0	0	1	1	0
1	0	1	0	1	0	1	0	0
1	1	0	0	1	0	0	1	0
1	1	1	1	0	0	0	0	0